

NinaivX — Aims, Objectives, Research Questions & Contribution

Ashwin Kumar Ramesh Kumar · MSc Artificial Intelligence, University of Sheffield · Project Framing (draft)

1. Problem and motivation

Grief and loneliness are widespread and serious. The World Health Organization reports that around one in six people worldwide is affected by loneliness, linked to an estimated 871,000 deaths per year [WHO 2025]. When a loved one dies, the bereaved are left with photographs but no way to hear the person's voice or interact with their personality. In parallel, commercial “digital afterlife” services have begun to appear, demonstrating both genuine demand and significant ethical risk. This project investigates how a voice-based conversational system for the bereaved and the lonely can be designed to be *emotionally supportive*, *factually safe*, and *privacy-respecting*, and evaluates whether such a design is effective.

2. Aim

To design, implement, and evaluate a privacy-first, dual-mode voice conversational AI that (a) supports healthy grieving and companionship, and (b) reduces the risk of factual fabrication when representing a real, deceased person, through an explicit safety-oriented architecture — and to critically assess this approach against existing commercial systems and the relevant ethical literature.

3. Objectives

1. Review the literature on conversational AI, speech synthesis and voice cloning, AI companionship, the digital afterlife industry, and AI safety/ethics, to identify the gap (completed; see Literature Survey).
2. Design a dual-mode architecture comprising an *air-gapped*, *retrieval-constrained* “legacy” (deceased) mode and a *web-connected* “companion” mode, mediated by a supervisor router.
3. Implement the backend: agent orchestration, speech-to-text and text-to-speech, persona memory, voice cloning, and secure authentication (completed).
4. Evaluate the design through a **technical evaluation** of factual fabrication in the legacy (air-gapped) mode versus the companion (web-connected) mode, using **synthetic personas**. No human participants are involved (no user study), so no ethical approval is required.
5. Critically discuss the ethical implications — consent, transparency, and grief support versus grief replacement — relative to existing commercial systems.

4. Research questions

RQ1. Can an air-gapped, retrieval-constrained design reduce factual fabrication in a voice-based legacy persona compared with an unconstrained generative approach, while maintaining perceived conversational quality?

RQ2. How can a single dual-mode architecture satisfy the differing safety and capability requirements of a deceased (legacy) persona and a living (companion) persona?

5. Contribution

- A reproducible, openly-documented **dual-mode architecture** for safety-aware conversational voice AI.
- An explicit **safety- and privacy-oriented design** (air-gapping the deceased persona) that operationalises harms previously raised only in the ethics literature.
- An **empirical evaluation** of that safety design — the element commercial products do not publish.
- A **critical positioning** of the system against existing commercial offerings, including an ethical discussion of grief support versus grief replacement.

6. Positioning against existing systems (esp. You Only Virtual)

The existence of commercial systems such as You Only Virtual (YOV), HereAfter AI, Replika, and Project December confirms the demand for this technology; it does not diminish the academic value of this work, which lies in an explicit, open, evaluated, and ethically-grounded design rather than in commercial novelty. The clearest contrast is with YOV, whose founder publicly frames the goal as the *elimination* of grief; this project takes the opposite, evidence-informed position of *supporting* healthy grieving.

Dimension	You Only Virtual	NinaivX (this project)
Stance on grief	Aims to eliminate grief / avoid the “hopelessness of forever” (founder's stated goal)	Supports healthy grieving and closure; a comfort tool, not a permanent replacement
Personas	Deceased only	Dual-mode: deceased (legacy) and living companion
Factual safety	Generative; safety mechanisms not publicly documented	Explicit air-gapped, retrieval-constrained legacy mode to limit fabrication
Connectivity	Not publicly specified	Mode-dependent: legacy air-gapped, companion web-connected

Privacy	US commercial product	Privacy-first design (air-gapping; data-sovereignty intent)
Openness	Closed, proprietary	Open, documented, reproducible architecture
Evaluation	None published	Empirical evaluation of the safety design (core of this dissertation)

Transparency (implemented): the legacy persona discloses its artificial nature — it stays in character but honestly confirms it is an AI recreation if asked, and the app surfaces a disclosure notice (provided by the API) at onboarding and as an in-app indicator, in line with transparency recommendations in the ethics literature (Hollanek 2024).

Note: bracketed sources above refer to entries in the project's Literature Survey and reference list.